

Zero-Shot and Few-Shot Cross-Lingual Sentiment Analysis from English to Krio: A Comparative Study of mBERT and XLM-R

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Abstract

Krio, a low-resource English-lexified creole spoken primarily in Sierra Leone, lacks publicly available labelled datasets for most NLP tasks. This pilot study investigates cross-lingual transfer learning for ternary sentiment analysis (Negative, Neutral, Positive) from English to Krio using two multilingual transformers: mBERT and XLM-R. We manually annotated 200 Krio sentences sampled from the Jaward/Krio-Corpus and evaluated models in zero-shot, 10-shot, and 50-shot settings. The results show that few-shot learning significantly improves performance, with mBERT achieving a peak of 58.0% accuracy. Balanced sampling further increased Macro F1 to 0.324, although at the cost of raw accuracy. The error analysis highlights persistent bias toward the Neutral class and difficulties with narrative style, variable orthography, and culturally embedded expressions. We release the annotated Krio sentiment dataset to support future research on creole languages.

1 Introduction

Low-resource languages, such as Krio, pose challenges for NLP due to limited annotated data, inconsistent spelling, and grammatical differences from their lexifier languages. Krio (also known as Sierra Leone Creole) is an English-lexified creole spoken as a lingua franca by over five million people in Sierra Leone and its diaspora. It emerged in the late 18th and 19th centuries through contact between freed African slaves, local West African languages (such as Yoruba, Igbo, and Mende), and English. While Krio shares considerable lexical overlap with English, its syntax, tense-aspect system, and phonology make

it distinct, rendering direct application of English-trained models suboptimal.

Cross-lingual transfer leverages knowledge from high-resource languages such as English. This project compares two widely used multilingual models: mBERT (Devlin et al., 2019) and XLM-R (Conneau et al., 2020), on a 3-class sentiment analysis task for Krio.

Research Question: How effective are mBERT and XLM-R for zero-shot and few-shot cross-lingual sentiment transfer from English to Krio?

Hypothesis: XLM-R will outperform mBERT due to its larger and cleaner pre-training corpus and more robust tokenizer. Additionally, few-shot learning with 10–50 target language examples will substantially improve performance over zero-shot baselines, and balanced sampling will enhance Macro F1 by reducing class bias.

This work contributes one of the first annotated sentiment resources for Krio and provides practical insight into the strengths and limitations of current multilingual models for English-lexified creoles.

2 Related Work

2.1 Multilingual Language Models and Cross-Lingual Transfer

Multilingual pre-trained language models have revolutionised cross-lingual NLP. Devlin et al. (Devlin et al., 2019) introduced mBERT, trained on Wikipedia data from 104 languages using a shared WordPiece vocabulary. Although effective for zero-shot transfer (Pires et al., 2019), mBERT often underperforms on low-resource languages due to limited exposure and vocabulary dilution.

Conneau et al. (Conneau et al., 2020) proposed XLM-R, trained on a significantly larger CommonCrawl corpus covering 100 languages. Due to its scale and SentencePiece tokenization, XLM-R consistently outperforms mBERT, particularly in low-resource and zero-shot scenarios (Koto et al., 2024; Přibáň et al., 2024).

2.2 Cross-Lingual and Few-Shot Sentiment Analysis

Cross-lingual sentiment analysis has been approached through translation, label projection, and direct fine-tuning of multilingual models. Recent studies emphasise the value of few-shot learning with pretrained transformers (Winata et al., 2022). Class imbalance remains a major issue; techniques such as balanced sampling and class weighting have been shown to improve Macro F1 in low-data regimes (Ochal et al., 2023).

2.3 Sentiment Analysis for Creoles and African Languages

Creole languages combine lexical items from a dominant language (often English) with substrate grammars, resulting in unique syntactic and orthographic patterns. The NaijaSenti corpus (Muhammad et al., 2022) for Nigerian Pidgin demonstrated strong results using multilingual models and underscored the importance of native-speaker annotation. The CreoleVal benchmark (Häberle et al., 2024) further showed that while English lexical overlap helps, models still strug-

gle with syntactic divergence and spelling variation in creoles.

Very little prior work exists on Krio sentiment analysis. This project builds upon the parallel story corpus by Jaward (Jaward/Krio-Corpus) to create and publicly release a new annotated Krio sentiment dataset.

2.4 Gap in the Literature

While mBERT and XLM-R have been extensively studied, systematic comparisons in extremely low-resource creole settings with explicit focus on class imbalance and balanced few-shot learning remain scarce. This pilot addresses this gap.

2.5 English Source Dataset

Due to technical issues with Sentiment140, we used the ‘dair-ai/emotion’ dataset mapped to three sentiment classes: Negative (0), Neutral (1), and Positive (2). Around 4,000 examples were used for zero-shot training.

2.6 Krio Target Dataset

We sampled 200 sentences from the ‘Jaward/Krio-Corpus’ (937 parallel English-Krio sentences from 33 stories). Sentences were manually annotated by the author.

Annotation Guidelines

- Negative (0): Fear, anger, pain, disappointment, or danger.
- Neutral (1): Factual or descriptive narrative without strong emotional valence.
- Positive (2): Expressions of joy, gratitude, success, or positive emotion.

Borderline cases were resolved by considering overall story context.

Dataset Statistics:

- Average sentence length: 18.4 tokens
- Vocabulary size: 1,237 unique tokens
- Average words per sentence: 16.7

- Max words: 50

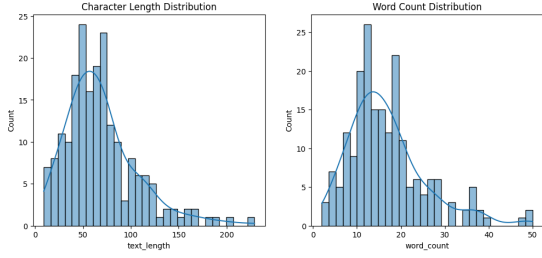


Figure 1: Character length and Word count distribution

- Label distribution: Heavily skewed toward Neutral (60%)

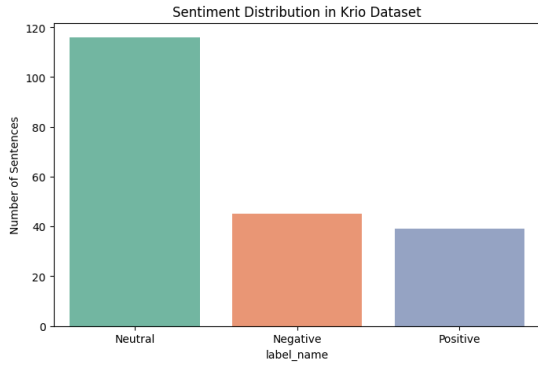


Figure 2: Label Distribution in Annotated Krio Dataset

Splits: 50 examples for few-shot training (with balanced sampling variant using 16 per class), and 150 examples for testing.

Preprocessing: Only basic cleaning while preserving original Krio spelling and orthographic variation.

3 Methodology

We fine-tune ‘bert-base-multilingual-cased’(mBERT) and ‘xlm-roberta-base’(XLM-R) using the Hugging Face ‘Trainer’ API. Experiments include:

- Zero-shot: Train only on English data.
- Few-shot (10-shot and 50-shot): Joint fine-tuning with small numbers of Krio examples.
- Balanced 50-shot: Equal number of examples per class via resampling.

- Baseline: TF-IDF vectorizer + Logistic Regression trained on English data.

Hyperparameters: Learning rate = $2e-5$, batch size = 8, max sequence length = 128, 3 epochs.

Evaluation metrics: Accuracy and Macro F1. All code was implemented in Colab notebooks for reproducibility.

4 Experiments and Results

The classical baseline (TF-IDF + Logistic Regression) achieved only 21.5% accuracy and 0.15 Macro F1, confirming the difficulty of traditional methods in cross-lingual settings.

4.1 Main Results

Experiment	mBERT	XLMR
Zero shot	0.200 / 0.168	0.180 / 0.169
10 shot	0.573 / 0.263	0.227 / 0.123
50 shot (unbalanced)	0.580 / 0.245	0.580 / 0.245
50 shot (balanced)	0.453 / 0.324	—

4.2 Unbalanced vs Balanced Few-Shot

Setting	Accu	Macro F1	Obser
50-shot (ub)	0.580	0.245	Neutral class bias
50-shot (b)	0.453	0.324	More balanced

Table 1: Unbalanced vs Balanced 50-shot (mBERT)

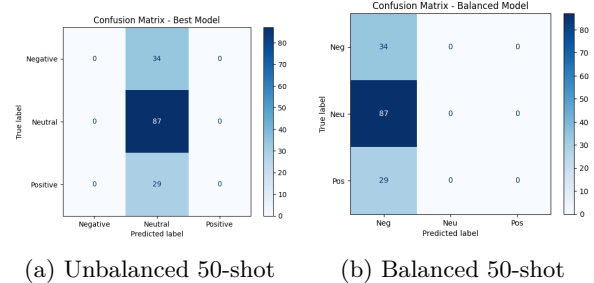


Figure 3: Confusion Matrices - Unbalanced vs Balanced

Interpretation: Few-shot learning brings clear gains over the zero-shot baseline. Balanced sampling trades some accuracy for substantially better Macro F1,

showing improved fairness across classes and a reduction in the strong bias toward the Neutral category. XLM-R did not outperform mBERT as hypothesised, likely due to the very small data scale and differences in how the two models handle Krio’s variable orthography.

5 Error Analysis

Common failure patterns include:

- Bias toward the Neutral class The model frequently defaults to Neutral predictions, especially for longer or more complex sentences. Even after balanced sampling, Neutral bias persists, though to a lesser degree.
- Variable Orthography and Tokenization Issues Krio’s non-standardized spelling significantly affects subword tokenization in both mBERT and XLM-R.
- Culturally Embedded Expressions Expressions such as “a beg”, “waiooo”, “debul”, and “krōkrō” carry strong emotional connotations that the model struggles to interpret without cultural context.

Table 2: Error Rate by Linguistic Feature

Feature	Count	ER(%)	Accu(%)
Story Description	124	41.94	58.06
Movement/Action	46	43.48	56.52
Positive Emotion	6	66.67	33.33
Fear/Pleading	3	100.00	0.00
Rash/Pain	1	100.00	0.00
Devil/Danger	1	0.00	100.00
Greed	1	0.00	100.00

Miscalssified Examples

- Text: “Papa na mi krōkrō de krach mi, duya a beg di wata we den kin tek fō was mi...” True: Negative → Predicted: Neutral
Strong pain/pleading expression missed by the model.

- Text: “Di pus de laf am...”
True: Positive → Predicted: Neutral
Clear positive emotion (“laf” = laugh) not captured.
- Text: “As i wan chap di sista so nain di bōbō ala, ‘waiooo mi krōkrō oo...’”
True: Negative → Predicted: Neutral
Intensifier + culturally loaded exclamation ignored.

6 Discussion

This pilot study demonstrates that multilingual transformers can achieve reasonable performance on ternary sentiment analysis for Krio, reaching up to 58% accuracy with as few as 50 labeled examples. The substantial improvement from few-shot learning (from approximately 20% in zero-shot to 58%) validates the effectiveness of cross-lingual transfer for English-lexified creoles, even in extremely low-resource settings.

Contrary to the hypothesis, mBERT slightly outperformed XLM-R in the few-shot setting. We believe this is mainly due to the very small data scale and XLM-R’s larger vocabulary, which likely caused more tokenization issues with Krio’s non-standard orthography. Balanced sampling successfully reduced the strong bias toward the Neutral class and improved Macro F1 from 0.245 to 0.324, though at the cost of raw accuracy, a common trade-off in imbalanced classification tasks.

6.1 Key Insights

- Lexical similarity to English offers a helpful foundation, but it is not enough to overcome grammatical differences and orthographic variation.
- Culturally specific expressions and narrative-style sentences remain particularly challenging for the models.
- Class imbalance continues to be a major issue, even when using resampling techniques.

7 Limitations and Future Work

7.1 Limitations

- Small dataset size: Only 200 annotated sentences (150 for testing) limits statistical significance and generalizability.
- Single annotator: All annotations were performed by the author, which may introduce some subjectivity.
- Limited experimentation: We did not explore continued pre-training, language-specific adapters (such as MAD-X or LoRA), or data augmentation techniques.
- Domain mismatch: The Jaward corpus primarily contains narrative stories, which may not fully represent everyday spoken or social media Krio.
- Compute constraints: Experiments were restricted to base models due to limited computational resources.

7.2 Future Work

- Significantly expand the annotated Krio sentiment dataset
- Include more diverse data such as code-switched and spontaneous spoken Krio.
- Explore human-in-the-loop active learning to grow the dataset more efficiently.
- Experiment with adapter-based methods and larger models (e.g., XLM-R Large).

8 Conclusion

This pilot study demonstrates the viability of cross-lingual transfer for ternary sentiment analysis in Krio, a low-resource English-lexified creole. Using mBERT and XLM-R, we showed that few-shot learning can substantially improve performance, raising accuracy from around 20% in the zero-shot setting to 58% with only 50 annotated examples.

The results confirm the effectiveness of few-shot fine-tuning and balanced sampling, while also highlighting persistent challenges such as variable orthography, narrative style, and culturally embedded expressions. Notably, mBERT outperformed XLM-R in this extremely low-resource scenario, underscoring the need for careful empirical testing of multilingual models.

By manually annotating and publicly releasing a 200-sentence Krio sentiment dataset, this project makes a concrete contribution to Krio NLP resources. Overall, the findings affirm that meaningful progress on low-resource creoles is possible even with limited labeled data.

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